



LAB EXPERIMENTS

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SUBJECT: Agentic AI

Experiment 1: Text-to-SQL Workflow

Objective:

To develop a Python program that accepts a natural language question, generates an SQL query using an LLM, executes the query on a SQLite database, and displays the result.

Python Code:

```
import sqlite3

from ollama import chat

MODEL = "llama3.2"

db = sqlite3.connect("company.db")
cursor = db.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS employees (
    id INTEGER,
    name TEXT,
    department TEXT,
    salary INTEGER
)
""")

cursor.execute("SELECT COUNT(*) FROM employees")

if cursor.fetchone()[0] == 0:
    data = [
        (1, "John", "IT", 60000),
        (2, "Alice", "HR", 50000),
        (3, "Bob", "IT", 70000),
        (4, "David", "Sales", 55000),
        (5, "Emma", "Finance", 65000)
    ]
```

```
cursor.executemany(
    "INSERT INTO employees VALUES (?, ?, ?, ?)",
    data
)
```

```
db.commit()
```

```
question = input("Enter your question: ")
```

```
prompt = f"""
```

Convert the following question into a SQLite SELECT query.

Database:

```
employees(id, name, department, salary)
```

Question:

```
{question}
```

Return only the SQL query.

```
"""
```

```
response = chat(
    model=MODEL,
    messages=[
        {"role": "user", "content": prompt}
    ]
)
```

```
sql = response.message.content.strip()
```

```
sql = sql.replace("`sql", "").replace("`", "").strip()
```

```
print("\n===== GENERATED SQL =====")
print(sql)
```

```
if sql.upper().startswith("SELECT"):
```

```
    cursor.execute(sql)
```

```
    result = cursor.fetchall()
```

```
    print("\n===== RESULT =====")
```

```
    for row in result:
```

```
        print(row)
```

```
else:
```

```
    print("Only SELECT queries are allowed.")
```

```
db.close()
```

Input:

Enter your question: Show all employees in IT

Output:

```
PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteS
d) ; (& "c:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments\.venv\Scripts\Activate.ps1")
(.venv) PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> python experiment1_text_to_sql.py
Ask a question: Show all employees in IT

Generated SQL:
SELECT * FROM employees WHERE department = 'IT';

Database Result:
(1, 'John', 'IT', 60000)
(3, 'Bob', 'IT', 70000)
[]
```

Experiment 2: RAG-Based Question Answering System

Objective:

To implement a basic Retrieval-Augmented Generation system that indexes documents, retrieves relevant information, and generates an answer using an LLM.

Python Code:

```
from ollama import chat
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
MODEL = "llama3.2"
```

```
with open("sample.txt", "r", encoding="utf-8") as file:
```

```
    text = file.read()
```

```
documents = [
```

```
    x.strip()
```

```
    for x in text.split("\n\n")
```

```
    if x.strip()
```

```
]
```

```
vectorizer = TfidfVectorizer()
```

```
vectors = vectorizer.fit_transform(documents)
```

```
question = input("Enter your question: ")
```

```
query_vector = vectorizer.transform([question])
```

```
scores = cosine_similarity(
```

```
    query_vector,
```

```
    vectors
```

```
)[0]
```

```
top_indices = scores.argsort()[-3:][::-1]
```

```
context = ""
```

```
for index in top_indices:
```

```
    context += documents[index] + "\n"
```

```
prompt = f"""
```

Answer the question using only the following context.

Context:

```
{context}
```

Question:

```
{question}
```

Give a simple and accurate answer.

```
"""
```

```
response = chat(  
    model=MODEL,  
    messages=[  
        {"role": "user", "content": prompt}  
    ]  
)
```

```
print("\n===== RETRIEVED INFORMATION =====")
```

```
print(context)
```

```
print("\n===== RESPONSE =====")
```

```
print(response.message.content)
```

sample.txt:

Artificial Intelligence is a branch of computer science that focuses on creating intelligent systems.

Machine Learning is a subset of Artificial Intelligence that allows computers to learn from data.

Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers.

Cybersecurity protects computers, networks and data from cyber attacks.

Cloud computing provides computing resources through the internet.

Input:

Enter your question: What is Machine Learning?

Output:

```
PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments\.venv\Scripts\Activate.ps1")
(.venv) PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> python experiment2_rag_qa.py
Creating document index...
RAG system ready.

Ask a question: What is deep learning?

Retrieved Documents:
Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers.

Machine Learning is a subset of Artificial Intelligence. It allows computers to learn patterns from data and make predictions without being explicitly programmed for every task.

Cybersecurity is the practice of protecting computers, networks, applications and data from unauthorized access and cyber attacks.

Final Answer:
Deep Learning.
(.venv) PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments>
```

Experiment 3: Prompt Chaining for Summarization

Objective:

To implement a multi-step prompt pipeline where the LLM summarizes text, extracts important points, and generates questions from the extracted information.

Python Code:

```
from ollama import chat
```

```
MODEL = "llama3.2"
```

```
text = input("Enter your text: ")
```

```
# Step 1: Summarization
```

```
prompt1 = f"""
```

```
Summarize the following text in 5 simple sentences.
```

```
Text:
```

```
{text}
```

```
"""
```

```
response1 = chat(
```

```
model=MODEL,
messages=[
    {"role": "user", "content": prompt1}
]
)
```

```
summary = response1.message.content
```

```
# Step 2: Extract key points
```

```
prompt2 = f"""
```

```
Extract the 5 most important points from this summary:
```

```
{summary}
```

```
"""
```

```
response2 = chat(
```

```
    model=MODEL,
```

```
    messages=[
```

```
        {"role": "user", "content": prompt2}
```

```
    ]
```

```
)
```

```
key_points = response2.message.content
```

```
# Step 3: Generate questions
```

```
prompt3 = f"""
```

```
Create 3 questions based on these key points:
```

```
{key_points}
```

```
"""
```

```
response3 = chat(
```



```
model=MODEL,  
messages=[  
    {"role": "user", "content": prompt3}  
]  
)
```

```
questions = response3.message.content
```

```
print("\n===== SUMMARY =====")  
print(summary)
```

```
print("\n===== KEY POINTS =====")  
print(key_points)
```

```
print("\n===== QUESTIONS =====")  
print(questions)
```

Input:

Enter your text:

Artificial Intelligence is transforming many industries.

Machine learning helps computers learn from data.

Deep learning uses neural networks with multiple layers.

Generative AI can create text, images and code.

AI is widely used in healthcare, education and cybersecurity.

Output:

```

PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments\.venv\Scripts\Activate.ps1")
(.venv) PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> python experiment3_prompt_chaining.py
Enter text to summarize:
Artificial Intelligence is transforming many industries.

STEP 1 - SUMMARY
Here is a summary of the text in 5 simple sentences:

Artificial intelligence (AI) has become increasingly important. It is affecting various sectors and ways of life. AI technology continues to advance rapidly. The impact of AI can be seen across multiple fields, from business to healthcare. As AI develops further, it may have significant effects on different areas of society.

STEP 2 - KEY POINTS
Here are the 5 most important points extracted from the summary:

1. Artificial intelligence (AI) is becoming increasingly important and affecting various sectors and ways of life.
2. AI technology continues to advance rapidly, which may lead to significant effects on different areas of society.
3. The impact of AI can be seen across multiple fields, including business and healthcare.
4. As AI develops further, it will have a growing influence on society as a whole.
5. AI has the potential to have significant effects on different areas of society, but the extent of these effects is not yet fully understood.

STEP 3 - QUESTIONS
Here are three questions to test understanding based on the key points:

1. What is one way in which artificial intelligence (AI) is already affecting various sectors and ways of life?
(Answer should mention a specific area, such as business or healthcare.)

2. According to point 3, where can the impact of AI be seen across multiple fields?
(Answer should mention "business and healthcare", or a similar combination.)

3. What is one potential consequence of AI developing further, according to point 4?
(Answer should mention something like "a growing influence on society as a whole", or another relevant phrase.)

PROMPT CHAIN COMPLETED

```

Experiment 4: SQL Agent with Tool Use

Objective:

To develop a SQL agent that uses an LLM and a database tool to understand user questions, execute SQL queries, and generate the final response.

Python Code:

```
import sqlite3
```

```
from ollama import chat
```

```
MODEL = "llama3.2"
```

```
db = sqlite3.connect("sales.db")
```

```
cursor = db.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS sales (
```

```
    id INTEGER,
```

```
        product TEXT,  
        quantity INTEGER,  
        price INTEGER  
    )  
""")  
  
cursor.execute("SELECT COUNT(*) FROM sales")
```

```
if cursor.fetchone()[0] == 0:  
    data = [  
        (1, "Laptop", 5, 50000),  
        (2, "Phone", 10, 20000),  
        (3, "Tablet", 7, 15000),  
        (4, "Monitor", 8, 12000),  
        (5, "Keyboard", 20, 2000)  
    ]
```

```
    cursor.executemany(  
        "INSERT INTO sales VALUES (?, ?, ?, ?)",  
        data  
    )
```

```
db.commit()
```

```
def run_sql(sql):  
    if not sql.strip().upper().startswith("SELECT"):  
        return "Only SELECT queries are allowed."  
  
    try:  
        cursor.execute(sql)  
        return str(cursor.fetchall())
```

```
except Exception as e:
```

```
    return str(e)
```

```
question = input("Enter your question: ")
```

```
prompt = f"""
```

```
You are a SQL database agent.
```

```
Database:
```

```
sales(id, product, quantity, price)
```

```
User question:
```

```
{question}
```

```
Generate only a SQLite SELECT query.
```

```
"""
```

```
response = chat(
```

```
    model=MODEL,
```

```
    messages=[
```

```
        {"role": "user", "content": prompt}
```

```
    ]
```

```
)
```

```
sql = response.message.content.strip()
```

```
sql = sql.replace("`sql", "").replace("`", "").strip()
```

```
print("\n===== AGENT GENERATED SQL =====")
```

```
print(sql)
```

```
result = run_sql(sql)
```

```
print("\n===== TOOL RESULT =====")
print(result)
```

```
final_prompt = f"""
User question:
{question}
```

```
Database result:
{result}
```

```
Give the final answer in simple English.
"""
```

```
final_response = chat(
    model=MODEL,
    messages=[
        {"role": "user", "content": final_prompt}
    ]
)
```

```
print("\n===== FINAL RESPONSE =====")
print(final_response.message.content)
```

```
db.close()
```

Input:

Enter your question: Which product has the highest quantity?

Output:

```

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• (.venv) PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> python experiment4_sql_agent.py
Ask a question about sales: Which product has the highest quantity?

Agent is working...

Tool: run_sql
Arguments: {'sql': 'SELECT product FROM sales WHERE quantity = (SELECT MAX(quantity) FROM sales)'}
Tool Result: [('Keyboard',)]

Final Answer:
The product with the highest quantity in the sales database is 'Keyboard'.
❖ (.venv) PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments>

```

Experiment 5: Multi-Agent SDR System

Objective:

To design a multi-agent system consisting of specialized agents for lead generation, lead qualification, and email generation.

Python Code:

```
from ollama import chat
```

```
MODEL = "llama3.2"
```

```
def lead_generation(industry):
```

```
    prompt = f"""
```

You are a Lead Generation Agent.

Generate 5 example B2B companies in the {industry} industry.

For each company provide:

1. Company name
2. Industry
3. Business type
4. Possible business need

Use fictional companies for demonstration.

```
"""
```

```
response = chat(  
    model=MODEL,  
    messages=[  
        {"role": "user", "content": prompt}  
    ]  
)
```

```
return response.message.content
```

```
def lead_qualification(leads):
```

```
    prompt = f"""
```

```
You are a Lead Qualification Agent.
```

```
Analyze the following leads:
```

```
{leads}
```

```
For each lead provide:
```

1. Company
2. Score from 1 to 10
3. Reason
4. Priority: HIGH, MEDIUM or LOW

```
"""
```

```
response = chat(  
    model=MODEL,  
    messages=[  
        {"role": "user", "content": prompt}  
    ]  
)
```

```
return response.message.content
```

```
def email_generation(qualified_leads):
```

```
    prompt = f"""
```

```
You are an Email Generation Agent.
```

```
Create professional email templates for
```

```
HIGH priority leads from the following list:
```

```
{qualified_leads}
```

```
Rules:
```

- Keep emails short.
- Keep them professional.
- Do not invent personal information.
- Do not actually send emails.

```
"""
```

```
response = chat(
```

```
    model=MODEL,
```

```
    messages=[
```

```
        {"role": "user", "content": prompt}
```

```
    ]
```

```
)
```

```
return response.message.content
```

```
industry = input("Enter target industry: ")
```



```
print("\n===== AGENT 1: LEAD GENERATION =====")
```

```
leads = lead_generation(industry)
```

```
print(leads)
```

```
print("\n===== AGENT 2: LEAD QUALIFICATION =====")
```

```
qualified = lead_qualification(leads)
```

```
print(qualified)
```

```
print("\n===== AGENT 3: EMAIL GENERATION =====")
```

```
emails = email_generation(qualified)
```

```
print(emails)
```

```
print("\n===== MULTI-AGENT WORKFLOW COMPLETED =====")
```

Input:

Enter target industry: AI

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments\.venv\Scripts\Activate.ps1")
(.venv) PS C:\Users\vodap\OneDrive\Pictures\Desktop\agentic ai experiments> python experiment5_multi_agent_sdr.py

=====
MULTI-AGENT SDR SYSTEM
=====

Enter target industry: Cybersecurity

[AGENT 1] Lead Generation Agent
-----
As a Lead Generation Agent, I've identified 5 potential B2B companies in the Cybersecurity industry that could benefit from our services:

1. **Company Name:** NovaTech Innovations
**Industry:** Renewable Energy
**Company Type:** Manufacturing
**Possible Business Need:** Threat Intelligence Solution for Industrial Control Systems (ICS)

NovaTech Innovations is a leading manufacturer of solar panel systems and energy storage solutions. With the increasing demand for renewable energy, they are now exploring ways to secure their industrial control systems from cyber threats. A threat intelligence solution could help them identify potential vulnerabilities and prevent attacks on their ICS.

2. **Company Name:** MedCorp Solutions
**Industry:** Healthcare Technology
**Company Type:** Software Development
**Possible Business Need:** Endpoint Security Solution for Medical Device Integration

MedCorp Solutions is a software development company that specializes in creating medical device integration solutions for healthcare providers. With the increasing use of medical devices, they are now concerned about protecting sensitive patient data from cyber threats. An endpoint security solution could help them ensure the integrity and confidentiality of their customers' data.

Start JSON Server Ln 123, Col 11 Spaces: 4
```